

GATE Data Science & Artificial Intelligence (DA)

Comprehensive 80-Question Topic-Wise Question Bank

10 PYQs + 10 Practice Questions with Detailed Step-by-Step Solutions per Topic

Topics: Sampling & Estimation, Support Vector Machines, BCNF & Decompositions, Mensuration & Geometry

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1 Topic 1: Sampling & Estimation Theory

1.1 10 Previous Year Questions (PYQs)

Q1.1

[GATE DA 2024 | NAT | 2 Marks]

Let $X_1, X_2, \dots, X_n$ be an independent and identically distributed (i.i.d.) random sample from a normal distribution $\mathcal{N}(\mu, \sigma^2)$ where $\sigma^2 > 0$ is known. Consider the sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$. What is the minimum sample size n required so that the width of the 95% confidence interval for μ is at most 0.5σ ? (Use standard normal quantile $z_{0.025} = 1.96$) **Answer: 62** **Solution:** The 95% confidence interval is $\left[\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right]$. The width is $W = 2z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 2(1.96) \frac{\sigma}{\sqrt{n}} = \frac{3.92\sigma}{\sqrt{n}}$. Setting $W \leq 0.5\sigma \implies \frac{3.92}{\sqrt{n}} \leq 0.5 \implies \sqrt{n} \geq \frac{3.92}{0.5} = 7.84 \implies n \geq (7.84)^2 = 61.4656$. Thus, the minimum integer sample size is $n = 62$.

Q1.2

[GATE ST 2023 | MCQ | 1 Mark]

Let $X_1, X_2, \dots, X_n$ be an i.i.d. sample from a Poisson distribution with parameter $\lambda > 0$. The maximum likelihood estimator (MLE) of $\theta = e^{-\lambda} = P(X_1 = 0)$ is:

- (A) $e^{-\bar{X}}$
- (B) $\frac{1}{n} \sum_{i=1}^n \mathbb{I}(X_i = 0)$
- (C) $\bar{X}e^{-\bar{X}}$
- (D) $(1 - \frac{1}{n})^{\sum X_i}$

Answer: (A) **Solution:** The log-likelihood is $\ln L(\lambda) = \sum_{i=1}^n (x_i \ln \lambda - \lambda - \ln(x_i!))$. Maximizing yields $\hat{\lambda}_{MLE} = \bar{X}$. By the functional invariance property of MLEs, the MLE of any continuous function $g(\lambda)$ is $g(\hat{\lambda}_{MLE})$. Therefore, $\hat{\theta}_{MLE} = e^{-\bar{X}}$.

Q1.3

[GATE DA 2025 | MSQ | 2 Marks]

Let $X_1, X_2, \dots, X_n$ be an i.i.d. random sample from a distribution with finite population mean μ and finite population variance σ^2 . Let $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ and $S_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$. Which of the following statements is/are TRUE?

- (A) S^2 is an unbiased estimator of σ^2 .
- (B) S_n^2 is an asymptotically unbiased and consistent estimator of σ^2 .
- (C) $\mathbb{E}[S_n^2] = \frac{n-1}{n} \sigma^2$.
- (D) $\bar{X}$ is an unbiased and consistent estimator of μ .

Answer: (A, B, C, D) **Solution:** All four are classic properties. (A) Bessel's correction makes S^2 unbiased: $\mathbb{E}[S^2] = \sigma^2$. (C) $\mathbb{E}[S_n^2] = \frac{n-1}{n} \sigma^2 < \sigma^2$. (B) As $n \rightarrow \infty$, $\frac{n-1}{n} \rightarrow 1$, making S_n^2 asymptotically unbiased and consistent by the Weak Law of Large Numbers. (D) $\mathbb{E}[\bar{X}] = \mu$ and $\text{Var}(\bar{X}) = \frac{\sigma^2}{n} \rightarrow 0$.

Q1.4

[GATE ST 2022 | NAT | 2 Marks]

Let $X_1, X_2, \dots, X_n$ be a random sample from a uniform distribution $U(0, \theta)$, where $\theta > 0$. Let $X_{(n)} = \max(X_1, \dots, X_n)$. If $c \cdot X_{(n)}$ is an unbiased estimator of θ for $n = 9$, what is the value of c ? (Round off to two decimal places) **Answer: 1.11** **Solution:** The cumulative distribution function of $X_{(n)}$ is $F_{(n)}(x) = \left(\frac{x}{\theta}\right)^n$, and its probability density function is $f_{(n)}(x) = \frac{nx^{n-1}}{\theta^n}$ for $0 \leq x \leq \theta$. Thus, $\mathbb{E}[X_{(n)}] = \int_0^\theta x \cdot \frac{nx^{n-1}}{\theta^n} dx = \frac{n}{\theta^n} \left[\frac{x^{n+1}}{n+1}\right]_0^\theta = \frac{n}{n+1} \theta$. For unbiasedness, $\mathbb{E}[cX_{(n)}] = c \frac{n}{n+1} \theta = \theta \implies c = \frac{n+1}{n}$. For $n = 9$, $c = \frac{10}{9} \approx 1.11$.

Q1.5

[GATE DA 2024 | MCQ | 1 Mark]

An estimator T_n of a parameter θ is said to be consistent if, for any arbitrarily small $\epsilon > 0$:

- (A) $\lim_{n \rightarrow \infty} P(|T_n - \theta| < \epsilon) = 1$
 (B) $\mathbb{E}[T_n] = \theta$ for all finite n
 (C) $\lim_{n \rightarrow \infty} \text{Var}(T_n) = 1$
 (D) $P(T_n = \theta) = 1$ for all n

Answer: (A) **Solution:** Consistency is defined by convergence in probability: $\lim_{n \rightarrow \infty} P(|T_n - \theta| \geq \epsilon) = 0$, which is equivalent to $\lim_{n \rightarrow \infty} P(|T_n - \theta| < \epsilon) = 1$.

Q1.6

[GATE ST 2021 | NAT | 2 Marks]

Let $X_1, X_2, \dots, X_5$ be an i.i.d. sample from an exponential distribution with probability density function $f(x; \theta) = \theta e^{-\theta x}$ for $x > 0$ and $\theta > 0$. The observed values are 1.2, 0.8, 1.5, 0.5, 1.0. What is the maximum likelihood estimate of θ ? **Answer: 1.00** **Solution:** The log-likelihood is $\ln L(\theta) = n \ln \theta - \theta \sum_{i=1}^n x_i$. Setting the derivative to zero: $\frac{n}{\theta} - \sum_{i=1}^n x_i = 0 \implies \hat{\theta}_{\text{MLE}} = \frac{n}{\sum_{i=1}^n x_i} = \frac{1}{\bar{x}}$. The sample sum is $1.2 + 0.8 + 1.5 + 0.5 + 1.0 = 5.0$. Thus, $\hat{\theta} = \frac{5}{5.0} = 1.00$.

Q1.7

[GATE DA 2025 | MCQ | 2 Marks]

Let $X_1, X_2, \dots, X_n$ be an i.i.d. sample from a normal distribution $\mathcal{N}(\mu, 1)$. The Cramér-Rao Lower Bound (CRLB) for the variance of any unbiased estimator of μ is:

- (A) 1
 (B) $\frac{1}{n}$
 (C) $\frac{1}{n^2}$
 (D) $\frac{\mu}{n}$

Answer: (B) **Solution:** The log-likelihood for a single observation is $\ln f(x; \mu) = -\frac{1}{2} \ln(2\pi) - \frac{1}{2}(x - \mu)^2$. Then $\frac{\partial \ln f}{\partial \mu} = (x - \mu)$ and $\frac{\partial^2 \ln f}{\partial \mu^2} = -1$. The Fisher Information for a single observation is $I_1(\mu) = -\mathbb{E} \left[\frac{\partial^2 \ln f}{\partial \mu^2} \right] = 1$. For a sample of size n , $I_n(\mu) = nI_1(\mu) = n$. The CRLB is $\frac{1}{I_n(\mu)} = \frac{1}{n}$.

Q1.8

[GATE ST 2020 | NAT | 1 Mark]

Let X_1, X_2, X_3 be an i.i.d. sample from a Bernoulli distribution with parameter $p \in (0, 1)$. Consider two unbiased estimators of p : $T_1 = \frac{X_1 + X_2 + X_3}{3}$ and $T_2 = \frac{X_1 + 2X_2}{3}$. If the relative efficiency of T_2 with respect to T_1 is defined as $\frac{\text{Var}(T_1)}{\text{Var}(T_2)}$, what is this ratio? (Round off to two decimal places) **Answer: 0.60** **Solution:** $\text{Var}(X_i) = p(1 - p)$. Then $\text{Var}(T_1) = \frac{1}{9}(\text{Var}(X_1) + \text{Var}(X_2) + \text{Var}(X_3)) = \frac{3p(1-p)}{9} = \frac{p(1-p)}{3}$. Next, $\text{Var}(T_2) = \frac{1}{9}(\text{Var}(X_1) + 4\text{Var}(X_2)) = \frac{5p(1-p)}{9}$. The relative efficiency is $\frac{\text{Var}(T_1)}{\text{Var}(T_2)} = \frac{p(1-p)/3}{5p(1-p)/9} = \frac{1/3}{5/9} = \frac{3}{5} = 0.60$.

Q1.9

[GATE DA 2026 | MSQ | 2 Marks]

Which of the following statements is/are TRUE regarding Maximum Likelihood Estimators (MLE)?

- (A) Under standard regularity conditions, an MLE is asymptotically normally distributed.
 (B) An MLE is always unbiased for all sample sizes $n \geq 1$.
 (C) If a single sufficient statistic exists, the MLE is a function of that sufficient statistic.
 (D) The MLE is invariant under one-to-one transformations of the parameter.

Answer: (A, C, D) **Solution:** (A) Asymptotic normality is a standard property: $\sqrt{n}(\hat{\theta}_{\text{MLE}} - \theta) \xrightarrow{d} \mathcal{N}(0, I_1(\theta)^{-1})$. (B) is false; e.g., the MLE of σ^2 in a normal distribution is $\frac{1}{n} \sum (X_i - \bar{X})^2$, which is biased. (C) By the Neyman-Fisher factorization theorem, maximizing likelihood depends on the data only through the sufficient statistic. (D) Functional invariance holds for any transformation.

Q1.10**[GATE ST 2024 | NAT | 1 Mark]**

A random sample of size $n = 16$ from a normal distribution $\mathcal{N}(\mu, \sigma^2)$ gives sample mean $\bar{x} = 24$ and sample variance $s^2 = 16$. To test $H_0 : \mu = 20$ versus $H_1 : \mu \neq 20$, what is the computed value of the t -statistic? **Answer: 4.00** **Solution:** The test statistic is $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{24 - 20}{\sqrt{16}/\sqrt{16}} = \frac{4}{4/4} = \frac{4}{1} = 4.00$.

1.2 10 Practice Questions (GATE Level)**P1.1****[Practice | MCQ | 1 Mark]**

Let $X_1, X_2, \dots, X_n$ be an i.i.d. sample from a Rayleigh distribution with PDF $f(x; \sigma) = \frac{x}{\sigma^2} e^{-x^2/(2\sigma^2)}$ for $x \geq 0$. The MLE of σ^2 is:

- (A) $\frac{1}{2n} \sum_{i=1}^n X_i^2$
- (B) $\frac{1}{n} \sum_{i=1}^n X_i$
- (C) $\sqrt{\frac{1}{n} \sum_{i=1}^n X_i^2}$
- (D) $\frac{1}{n} \sum_{i=1}^n X_i^2$

Answer: (A) **Solution:** Log-likelihood: $\ln L(\sigma^2) = \sum \ln(x_i) - n \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum x_i^2$. Differentiating with respect to σ^2 and equating to zero: $-\frac{n}{\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum x_i^2 = 0 \implies \hat{\sigma}^2 = \frac{1}{2n} \sum_{i=1}^n X_i^2$.

P1.2**[Practice | NAT | 2 Marks]**

Let X_1, X_2, X_3 be an i.i.d. sample from a population with mean μ and variance σ^2 . An estimator of μ is proposed as $\hat{\mu} = aX_1 + 2aX_2 + 3aX_3$. What value of a makes $\hat{\mu}$ an unbiased estimator of μ ? (Round off to two decimal places) **Answer: 0.17** **Solution:** $\mathbb{E}[\hat{\mu}] = a\mu + 2a\mu + 3a\mu = 6a\mu$. For unbiasedness, $\mathbb{E}[\hat{\mu}] = \mu \implies 6a = 1 \implies a = \frac{1}{6} \approx 0.17$.

P1.3**[Practice | MSQ | 2 Marks]**

Let T_n be an estimator of θ computed from a sample of size n . Which of the following conditions is/are SUFFICIENT to guarantee that T_n is a consistent estimator of θ ?

- (A) $\lim_{n \rightarrow \infty} \mathbb{E}[T_n] = \theta$ and $\lim_{n \rightarrow \infty} \text{Var}(T_n) = 0$
- (B) $\lim_{n \rightarrow \infty} \text{MSE}(T_n) = 0$
- (C) T_n is unbiased for all n
- (D) $\lim_{n \rightarrow \infty} P(|T_n - \theta| \geq \epsilon) = 0$ for all $\epsilon > 0$

Answer: (A, B, D) **Solution:** (B) Mean squared error $\text{MSE}(T_n) = \text{Var}(T_n) + (\text{Bias}(T_n))^2$. By Chebyshev's inequality, $\text{MSE} \rightarrow 0$ implies convergence in probability (consistency). (A) satisfies this condition directly. (D) is the definition of consistency. (C) is not sufficient: e.g., $T_n = X_1$ is unbiased for all n , but its variance remains $\sigma^2 > 0$ and never vanishes.

P1.4**[Practice | NAT | 1 Mark]**

A random sample of 100 observations gives sample mean $\bar{x} = 50$. The population variance is known to be $\sigma^2 = 64$. What is the margin of error for a 95% confidence interval for the population mean? (Use $z_{0.025} = 1.96$) **Answer: 1.57** **Solution:** Margin of error $E = z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 1.96 \times \frac{\sqrt{64}}{\sqrt{100}} = 1.96 \times \frac{8}{10} = 1.96 \times 0.8 = 1.568 \approx 1.57$.

P1.5**[Practice | MCQ | 2 Marks]**

Let $X_1, X_2, \dots, X_n$ be i.i.d. observations from $U(\theta - 0.5, \theta + 0.5)$. Which of the following is TRUE about the maximum likelihood estimator of θ ?

- (A) It is unique and equals the sample median.

(B) It is not unique; any value in the interval $[X_{(n)} - 0.5, X_{(1)} + 0.5]$ is an MLE.

(C) It is strictly equal to $\frac{X_{(1)} + X_{(n)}}{2}$.

(D) The MLE does not exist.

Answer: (B) **Solution:** The joint likelihood is $L(\theta) = 1$ if $\theta - 0.5 \leq X_{(1)}$ and $X_{(n)} \leq \theta + 0.5$, which simplifies to $X_{(n)} - 0.5 \leq \theta \leq X_{(1)} + 0.5$, and 0 otherwise. Any θ inside this valid interval maximizes the likelihood.

P1.6

[Practice | NAT | 2 Marks]

Let $X \sim \text{Binomial}(n, p)$ where $n = 20$. If the observed value of X is 6, what is the maximum likelihood estimate of the odds ratio $\psi = \frac{p}{1-p}$? (Round off to two decimal places) **Answer: 0.43**

Solution: The MLE of p is $\hat{p} = \frac{X}{n} = \frac{6}{20} = 0.30$. By functional invariance of MLEs, $\hat{\psi} = \frac{\hat{p}}{1-\hat{p}} = \frac{0.30}{1-0.30} = \frac{0.30}{0.70} = \frac{3}{7} \approx 0.43$.

P1.7

[Practice | MCQ | 1 Mark]

By the Central Limit Theorem (CLT), the distribution of the sample mean $\bar{X}$ of n independent random variables with mean μ and variance $\sigma^2 < \infty$:

(A) Approaches a normal distribution as $n \rightarrow \infty$ regardless of the underlying population distribution.

(B) Is exactly normal for any finite n only if the population is uniformly distributed.

(C) Has variance that increases proportionally with n .

(D) Converges to the population median in distribution.

Answer: (A) **Solution:** The classical Lindeberg-Lévy CLT states that as sample size $n \rightarrow \infty$, the standardized variable $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$ converges in distribution to standard normal $\mathcal{N}(0, 1)$, irrespective of the underlying distribution shape.

P1.8

[Practice | NAT | 2 Marks]

Let $X_1, X_2, \dots, X_n$ be an i.i.d. sample from a normal distribution $\mathcal{N}(0, \sigma^2)$. The sample size is $n = 10$. The estimator $T = \frac{1}{c} \sum_{i=1}^{10} X_i^2$ is an unbiased estimator of σ^2 . What is the value of c ?

Answer: 10 **Solution:** $\mathbb{E}[X_i^2] = \text{Var}(X_i) + (\mathbb{E}[X_i])^2 = \sigma^2 + 0 = \sigma^2$. Thus, $\mathbb{E}\left[\sum_{i=1}^{10} X_i^2\right] = 10\sigma^2$. For T to be unbiased, $\mathbb{E}[T] = \frac{10\sigma^2}{c} = \sigma^2 \implies c = 10$.

P1.9

[Practice | MSQ | 2 Marks]

Which of the following statistics is/are sufficient for the specified parameter?

(A) $\sum_{i=1}^n X_i$ for parameter λ in an i.i.d. Poisson(λ) sample.

(B) $\max(X_1, \dots, X_n)$ for parameter θ in an i.i.d. $U(0, \theta)$ sample.

(C) $\sum_{i=1}^n X_i$ for parameter p in an i.i.d. Bernoulli(p) sample.

(D) The sample median for μ in an i.i.d. $\mathcal{N}(\mu, 1)$ sample.

Answer: (A, B, C) **Solution:** By the Factorization Criterion: (A) $\prod \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} = e^{-n\lambda} \lambda^{\sum x_i} \prod \frac{1}{x_i!}$. (B) $\prod \frac{1}{\theta} \mathbb{I}(x_i \leq \theta) = \frac{1}{\theta^n} \mathbb{I}(\max(x_i) \leq \theta)$. (C) Bernoulli joint PMF factors via $\sum x_i$. (D) is false; the sufficient statistic for normal mean is $\sum X_i$ (or $\bar{X}$), not the median.

P1.10

[Practice | NAT | 1 Mark]

A hypothesis test yields a test statistic with a two-tailed p -value equal to 0.032. At significance level $\alpha = 0.05$, is the null hypothesis rejected? Enter 1 for Yes and 0 for No. **Answer: 1** **Solution:** The standard decision rule states: if p -value $\leq \alpha$, reject H_0 . Here, $0.032 \leq 0.05$, so the null hypothesis is rejected.

2 Topic 2: Support Vector Machines – Margin & Kernel Trick

2.1 10 Previous Year Questions (PYQs)

Q2.1

[GATE DA 2024 | NAT | 2 Marks]

In a hard-margin Support Vector Machine for linearly separable binary classification, the separating hyperplane is given by $w_1x_1 + w_2x_2 + b = 0$. If the optimal weight vector is $w = [3, 4]^T$, what is the geometric margin (the total separation distance between the two supporting hyperplanes ± 1)? **Answer: 0.40** **Solution:** The canonical margin hyperplanes are $w^T x + b = +1$ and $w^T x + b = -1$. The geometric width of the margin is given by $\frac{2}{\|w\|_2}$. The Euclidean norm is $\|w\|_2 = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = 5$. Thus, the total margin is $\frac{2}{5} = 0.40$.

Q2.2

[GATE DA 2024 | MCQ | 1 Mark]

In a linear Support Vector Machine, removing a training data point that is NOT a support vector will cause the decision boundary to:

- (A) Shift toward the removed point.
- (B) Rotate by 90° .
- (C) Remain completely unchanged.
- (D) Collapse to the origin.

Answer: (C) **Solution:** In the dual formulation, the optimal weight vector is $w = \sum_{i=1}^n \alpha_i y_i x_i$. Non-support vectors have Lagrange multipliers $\alpha_i = 0$. Removing them has zero impact on the dual objective, leaving the decision boundary completely unchanged.

Q2.3

[GATE DA 2025 | MSQ | 2 Marks]

Which of the following functions $K(x, z)$ with $x, z \in \mathbb{R}^d$ satisfy Mercer's condition and define a mathematically valid SVM kernel?

- (A) $K(x, z) = (x^T z + 1)^2$
- (B) $K(x, z) = \exp(-\gamma \|x - z\|^2)$, where $\gamma > 0$
- (C) $K(x, z) = x^T z - 5$
- (D) $K(x, z) = (x^T z)^3$

Answer: (A, B, D) **Solution:** (A) is an inhomogeneous polynomial kernel. (B) is the Radial Basis Function (Gaussian RBF) kernel. (D) is a homogeneous polynomial kernel of degree 3. All three have symmetric, positive semi-definite Gram matrices. (C) is invalid because subtracting a positive constant can make the Gram matrix indefinite.

Q2.4

[GATE CS 2021 | NAT | 2 Marks]

Consider a 1-dimensional dataset with 3 points: $x_1 = 1$ with $y_1 = -1$, $x_2 = 2$ with $y_2 = +1$, and $x_3 = 4$ with $y_3 = +1$. Using a hard-margin linear SVM with decision function $f(x) = \text{sign}(wx + b)$, what is the optimal slope w ? **Answer: 2.00** **Solution:** The closest points of opposite classes are $x_1 = 1$ ($y_1 = -1$) and $x_2 = 2$ ($y_2 = +1$). These two points are the support vectors. The margin hyperplanes satisfy $w(1) + b = -1$ and $w(2) + b = +1$. Subtracting the two equations gives $w(2 - 1) = 1 - (-1) \implies w = 2.00$ (with $b = -3$). The boundary is at $x = 1.5$.

Q2.5

[GATE DA 2025 | MCQ | 1 Mark]

In soft-margin SVM, the objective function includes a regularization parameter $C > 0$: $\min_{w, b, \xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i$. As $C \rightarrow \infty$, the soft-margin SVM behavior approaches:

- (A) A hard-margin SVM that enforces zero training margin violations (if separable).

- (B) A model with an infinitely wide margin that misclassifies all points.
- (C) A standard unregularized logistic regression classifier.
- (D) A random coin-flip decision maker.

Answer: (A) **Solution:** As $C \rightarrow \infty$, the penalty for any slack $\xi_i > 0$ becomes infinite. The optimization algorithm is forced to set all $\xi_i = 0$, strictly enforcing $y_i(w^T x_i + b) \geq 1$, which is the exact hard-margin SVM formulation.

Q2.6 **[GATE CS 2022 | MCQ | 2 Marks]**

What is the primary computational advantage of using the “kernel trick” in Support Vector Machines?

- (A) It allows computing inner products in high-dimensional feature spaces implicitly without explicitly mapping data points to those spaces.
- (B) It eliminates the need for computing Lagrange multipliers during training.
- (C) It converts non-convex optimization problems into simple matrix inversions.
- (D) It reduces the number of support vectors strictly to 2.

Answer: (A) **Solution:** The kernel trick computes $K(x, z) = \langle \Phi(x), \Phi(z) \rangle$ directly in $O(d)$ input space operations, bypassing the expensive (or infinite-dimensional) construction of the explicit transformed vectors $\Phi(x)$.

Q2.7 **[GATE DA 2026 | NAT | 2 Marks]**

A dataset has 4 points in $\mathbb{R}^2$: $x_1 = (1, 1)$ ($y_1 = +1$), $x_2 = (1, -1)$ ($y_2 = +1$), $x_3 = (-1, 1)$ ($y_3 = -1$), and $x_4 = (-1, -1)$ ($y_4 = -1$). The optimal hard-margin SVM hyperplane has normal vector $w = [w_1, w_2]^T$ and bias b . What is the value of w_1 ? **Answer: 1.00** **Solution:** The points are separable by the vertical axis $x_1 = 0$. The support vectors are at $x_1 = +1$ (class +1) and $x_1 = -1$ (class -1). The margin condition gives $w_1(1) + b = 1$ and $w_1(-1) + b = -1$. Adding them gives $b = 0$, and subtracting gives $2w_1 = 2 \implies w_1 = 1.00$ (with $w_2 = 0$).

Q2.8 **[GATE CS 2023 | MSQ | 2 Marks]**

In the dual formulation of soft-margin SVM: $\max_{\alpha} \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j K(x_i, x_j)$ subject to $\sum_{i=1}^n \alpha_i y_i = 0$ and $0 \leq \alpha_i \leq C$. By the Karush-Kuhn-Tucker (KKT) complementary slackness conditions, which of the following is/are TRUE?

- (A) If $\alpha_i = 0$, the point x_i is correctly classified and lies outside or on the margin boundary ($\xi_i = 0$).
- (B) If $0 < \alpha_i < C$, the point x_i lies precisely on the margin boundary ($y_i f(x_i) = 1$ and $\xi_i = 0$).
- (C) If $\alpha_i = C$, the point x_i is a bounded support vector and may violate the margin ($\xi_i \geq 0$).
- (D) All support vectors must have $\alpha_i = C$.

Answer: (A, B, C) **Solution:** KKT conditions specify: (1) $\alpha_i = 0 \implies y_i f(x_i) > 1$ and $\xi_i = 0$. (2) $0 < \alpha_i < C \implies y_i f(x_i) = 1$ (free support vectors on the margin). (3) $\alpha_i = C \implies y_i f(x_i) \leq 1$ and $\xi_i \geq 0$ (bounded support vectors). (D) is false because free support vectors have $0 < \alpha_i < C$.

Q2.9 **[GATE DA 2024 | MCQ | 1 Mark]**

The feature space corresponding to the Gaussian RBF kernel $K(x, z) = \exp(-\gamma \|x - z\|^2)$ is:

- (A) Finite-dimensional with dimension equal to the input dimension d .
- (B) Finite-dimensional with dimension $\binom{d+2}{2}$.
- (C) Infinite-dimensional.

(D) 1-dimensional.

Answer: (C) **Solution:** Using Taylor series expansion: $\exp(2\gamma x^T z) = \sum_{k=0}^{\infty} \frac{(2\gamma)^k (x^T z)^k}{k!}$. The expansion contains terms of all polynomial degrees up to infinity, meaning the corresponding Hilbert reproducing kernel space is infinite-dimensional.

Q2.10

[GATE CS 2020 | NAT | 1 Mark]

If an SVM with Gaussian RBF kernel is trained with an extremely large value of γ ($\gamma \rightarrow \infty$), the model is highly prone to: enter 1 for Overfitting, or 0 for Underfitting. **Answer: 1** **Solution:** As $\gamma \rightarrow \infty$, the Gaussian kernel $K(x, z) \rightarrow 0$ for any distinct $x \neq z$. Each training example creates an isolated, narrow decision spike around itself, leading to severe overfitting (variance goes to maximum).

2.2 10 Practice Questions (GATE Level)

P2.1

[Practice | MCQ | 1 Mark]

In a hard-margin SVM, the optimization problem minimizes $\frac{1}{2}\|w\|^2$. Why is this quadratic objective preferred over minimizing $\|w\|$ directly?

- (A) $\frac{1}{2}\|w\|^2$ is strictly convex and differentiable everywhere, guaranteeing a unique global minimum.
- (B) Minimizing $\|w\|$ leads to non-linear constraints.
- (C) $\|w\|^2$ eliminates the dual formulation completely.
- (D) $\|w\|$ cannot be calculated in Euclidean space.

Answer: (A) **Solution:** The function $\frac{1}{2}\|w\|^2$ is a smooth, strictly convex quadratic function whose gradient is simply w , making it suitable for quadratic programming solvers without gradient singularities at $w = 0$.

P2.2

[Practice | NAT | 2 Marks]

Consider two support vectors $x_1 = [1, 2]^T$ with $y_1 = +1$ and $x_2 = [3, 4]^T$ with $y_2 = -1$ in a linear SVM. The optimal decision boundary is the perpendicular bisector of the segment connecting them. What is the Euclidean distance from the boundary to either support vector (the half-margin $\frac{1}{\|w\|}$)? (Round off to two decimal places) **Answer: 1.41** **Solution:** The vector between the two support vectors is $x_2 - x_1 = [3 - 1, 4 - 2]^T = [2, 2]^T$. The distance between them is $\|x_2 - x_1\|_2 = \sqrt{2^2 + 2^2} = \sqrt{8} = 2\sqrt{2}$. Since the hyperplane lies halfway between them, the margin distance to the boundary is $\frac{\sqrt{8}}{2} = \sqrt{2} \approx 1.41$.

P2.3

[Practice | MSQ | 2 Marks]

Let $K_1(x, z)$ and $K_2(x, z)$ be two valid Mercer kernels. Which of the following constructions also produce a valid Mercer kernel?

- (A) $K(x, z) = c_1 K_1(x, z) + c_2 K_2(x, z)$, where $c_1, c_2 > 0$
- (B) $K(x, z) = K_1(x, z) \cdot K_2(x, z)$
- (C) $K(x, z) = \exp(K_1(x, z))$
- (D) $K(x, z) = K_1(x, z) - K_2(x, z)$

Answer: (A, B, C) **Solution:** Kernels are closed under non-negative linear combinations (A), tensor/elementwise products by the Schur Product Theorem (B), and exponentiation (C). The difference of two positive semi-definite kernels (D) is not necessarily positive semi-definite.

P2.4

[Practice | NAT | 2 Marks]

For a homogeneous quadratic polynomial kernel $K(x, z) = (x^T z)^2$ with 2-dimensional inputs $x = [x_1, x_2]^T$ and $z = [z_1, z_2]^T$, the explicit feature mapping can be chosen as $\Phi(x) = [x_1^2, cx_1x_2, x_2^2]^T$. What is the value of $c > 0$? (Round off to two decimal places) **Answer: 1.41** **Solution:** $(x^T z)^2 = (x_1z_1 + x_2z_2)^2 = x_1^2z_1^2 + 2x_1x_2z_1z_2 + x_2^2z_2^2$. Taking inner product $\langle \Phi(x), \Phi(z) \rangle = x_1^2z_1^2 + c^2x_1x_2z_1z_2 + x_2^2z_2^2$. Matching coefficients gives $c^2 = 2 \implies c = \sqrt{2} \approx 1.41$.

P2.5

[Practice | MCQ | 1 Mark]

In a soft-margin SVM, what does a slack variable value $\xi_i = 1.5$ indicate for sample (x_i, y_i) ?

- (A) The point is correctly classified and lies on the margin.
- (B) The point is correctly classified and lies between the margin and the hyperplane.
- (C) The point is misclassified by the decision boundary.
- (D) The point has zero Lagrange multiplier.

Answer: (C) **Solution:** The constraint is $y_i(w^T x_i + b) \geq 1 - \xi_i$. If $\xi_i = 0$, the point is on or beyond the correct margin. If $0 < \xi_i < 1$, the point is correctly classified but inside the margin. If $\xi_i = 1$, the point lies exactly on the separating hyperplane. If $\xi_i > 1$, the point lies on the wrong side of the decision boundary (misclassified).

P2.6

[Practice | NAT | 2 Marks]

An SVM model finds 6 support vectors in a dataset of 100 training points. If the upper bound on the leave-one-out cross-validation error rate is given by $\frac{N_{SV}}{n}$, what is this bound? (Round off to two decimal places) **Answer: 0.06** **Solution:** By the Vapnik-Chervonenkis theorem for SVMs, $\mathbb{E}[P_{\text{error}}] \leq \frac{\mathbb{E}[N_{SV}]}{n}$. Here, $\frac{6}{100} = 0.06$.

P2.7

[Practice | MCQ | 2 Marks]

Consider mapping 1D points $x \in \mathbb{R}$ to a 2D space using $\Phi(x) = [x, x^2]^T$. The classes are: Class +1 at $x \in \{-2, 2\}$ and Class -1 at $x \in \{-1, 1\}$. In the transformed $\mathbb{R}^2$ space, these classes:

- (A) Remain linearly inseparable.
- (B) Are linearly separable by the horizontal line $\Phi_2(x) = 2.5$.
- (C) Require an RBF kernel to achieve separation.
- (D) Collapse to a single point.

Answer: (B) **Solution:** Mapped points: Class +1 at $(-2, 4)$ and $(2, 4)$, where $\Phi_2 = 4$. Class -1 at $(-1, 1)$ and $(1, 1)$, where $\Phi_2 = 1$. The horizontal line $\Phi_2 = 2.5$ (or $x^2 - 2.5 = 0$) separates them.

P2.8

[Practice | MSQ | 2 Marks]

Which of the following statements about Support Vector Regression (SVR) with ϵ -insensitive loss is/are TRUE?

- (A) Errors smaller than ϵ incur zero penalty in the loss function.
- (B) Points that lie strictly inside the ϵ -tube have Lagrange multipliers $\alpha_i = \alpha_i^* = 0$.
- (C) SVR solutions are sparse because only points on or outside the ϵ -tube act as support vectors.
- (D) The loss function grows quadratically for all errors beyond ϵ .

Answer: (A, B, C) **Solution:** ϵ -insensitive loss is $L_\epsilon(y, f(x)) = \max(0, |y - f(x)| - \epsilon)$. (A) Errors $\leq \epsilon$ have loss 0. (B) Points within the tube do not contribute to the dual solution ($\alpha = 0$). (C) This sparsity makes SVR efficient. (D) is false; standard ϵ -SVR uses a linear penalty for errors beyond ϵ .

P2.9

[Practice | NAT | 1 Mark]

A hard-margin SVM in $\mathbb{R}^3$ achieves a geometric margin of $\gamma = 0.25$. What is the Euclidean norm $\|w\|_2$ of its optimal weight vector? **Answer: 8.00** **Solution:** The geometric margin is $\gamma = \frac{2}{\|w\|_2}$. Setting $0.25 = \frac{2}{\|w\|_2} \implies \|w\|_2 = \frac{2}{0.25} = 8.00$.

P2.10

[Practice | MCQ | 1 Mark]

How does decreasing the regularization parameter C in a soft-margin SVM affect the bias-variance tradeoff?

- (A) Decreases bias, increases variance (risk of overfitting).
- (B) Increases bias, decreases variance (larger margin, more tolerance for violations).
- (C) Has no effect on bias or variance.
- (D) Both bias and variance decrease simultaneously.

Answer: (B) **Solution:** Decreasing C reduces the penalty on margin violations, placing more emphasis on maximizing margin width $\frac{2}{\|w\|}$. This leads to a smoother, simpler boundary with higher bias and lower variance.

3 Topic 3: Normalization – BCNF & Decomposition Properties

3.1 10 Previous Year Questions (PYQs)

Q3.1

[GATE CS 2023 | MCQ | 1 Mark]

A relation R with functional dependencies F is in Boyce-Codd Normal Form (BCNF) if and only if for every non-trivial functional dependency $X \rightarrow Y \in F^+$:

- (A) X is a superkey of R .
- (B) Y is a prime attribute of R .
- (C) X is a proper subset of some candidate key.
- (D) Y is a subset of X .

Answer: (A) **Solution:** BCNF requires that for every non-trivial functional dependency $X \rightarrow Y$, the determinant X must be a superkey of R .

Q3.2

[GATE CS 2021 | NAT | 2 Marks]

Consider relation $R(A, B, C)$ with functional dependencies:

$$F = \{AB \rightarrow C, C \rightarrow A\}$$

How many candidate keys does relation R have, and is R in BCNF? (Enter the number of candidate keys) **Answer: 2** **Solution:** Closures: $(AB)^+ = \{A, B, C\}$, so AB is a candidate key. Since $C \rightarrow A$, $(BC)^+ = \{B, C, A\}$, so BC is also a candidate key. Candidate keys are AB and BC . Prime attributes: $\{A, B, C\}$. In $C \rightarrow A$, C is not a superkey ($C^+ = \{A, C\} \neq \{A, B, C\}$), so R is not in BCNF. Number of candidate keys = 2.

Q3.3

[GATE DA 2024 | MSQ | 2 Marks]

Let a relation schema R be decomposed into R_1 and R_2 . The decomposition is a lossless-join decomposition with respect to a set of functional dependencies F if and only if:

- (A) $(R_1 \cap R_2) \rightarrow R_1 \in F^+$
- (B) $(R_1 \cap R_2) \rightarrow R_2 \in F^+$
- (C) $R_1 \cup R_2 = R$
- (D) $R_1 \cap R_2 = \emptyset$

Answer: (A, B, C) **Solution:** By the Lossless-Join Decomposition Theorem, $R = R_1 \cup R_2$ is lossless if and only if the common attributes $(R_1 \cap R_2)$ form a superkey of at least one of the decomposed schemas, i.e., $(R_1 \cap R_2) \rightarrow R_1$ or $(R_1 \cap R_2) \rightarrow R_2$. (D) results in a Cartesian product (lossy join).

Q3.4

[GATE CS 2020 | MCQ | 2 Marks]

Relation $R(A, B, C, D)$ has functional dependencies $F = \{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow A\}$. Which of the following statements is TRUE?

- (A) R is in 3NF but not BCNF.
- (B) R is in BCNF.
- (C) R is in 2NF but not 3NF.
- (D) R is in 1NF only.

Answer: (B) **Solution:** $A^+ = B^+ = C^+ = D^+ = \{A, B, C, D\}$. Every single attribute is a candidate key and therefore a superkey. Every determinant in F is a superkey, so R is in BCNF.

Q3.5

[GATE CS 2018 | MCQ | 2 Marks]

Consider the decomposition of relation $R(A, B, C)$ with $F = \{A \rightarrow B, B \rightarrow C, C \rightarrow A\}$ into $R_1(A, B)$ and $R_2(B, C)$. This decomposition is:

- (A) Lossless and dependency preserving.
- (B) Lossless but not dependency preserving.
- (C) Lossy but dependency preserving.
- (D) Neither lossless nor dependency preserving.

Answer: (A) **Solution:** $R_1 \cap R_2 = \{B\}$. In $R_2(B, C)$, $B \rightarrow C$ holds, meaning B is a key of R_2 , ensuring a lossless join. $A \rightarrow B$ is preserved in R_1 and $B \rightarrow C$ is preserved in R_2 . The closure of these covers $C \rightarrow A$. Thus, it is both lossless and dependency preserving.

Q3.6

[GATE CS 2019 | NAT | 2 Marks]

Relation $R(A, B, C, D, E)$ has functional dependencies $F = \{A \rightarrow B, BC \rightarrow D, D \rightarrow E\}$. If R is decomposed into BCNF schemas, what is the maximum number of relations obtained using the standard BCNF decomposition algorithm? **Answer: 3** **Solution:** Candidate key of R is AC . First violation: $A \rightarrow B$ (A is not superkey). Decompose into $R_1(A, B)$ (in BCNF) and $R_2(A, C, D, E)$ with dependencies $\{AC \rightarrow D, D \rightarrow E\}$. In R_2 , candidate key is AC , and $D \rightarrow E$ violates BCNF (D is not superkey). Decompose R_2 into $R_{2a}(D, E)$ (in BCNF) and $R_{2b}(A, C, D)$ (in BCNF, key AC). Total relations = 3.

Q3.7

[GATE DA 2025 | MSQ | 2 Marks]

Which of the following statements about normal forms and decompositions is/are TRUE?

- (A) Any relation schema can always be decomposed into a set of 3NF schemas that is both lossless and dependency preserving.
- (B) Any relation schema can always be decomposed into a set of BCNF schemas that is lossless.
- (C) Any relation schema can always be decomposed into a set of BCNF schemas that is both lossless and dependency preserving.
- (D) BCNF is strictly stronger than 3NF.

Answer: (A, B, D) **Solution:** (A) Bernstein's 3NF synthesis algorithm guarantees both lossless join and dependency preservation. (B) Standard BCNF decomposition guarantees losslessness, but dependency preservation cannot always be achieved (C is false). (D) BCNF does not permit the Y is prime exception allowed in 3NF.

Q3.8

[GATE CS 2022 | NAT | 1 Mark]

Relation $R(X, Y, Z)$ has dependencies $F = \{X \rightarrow Y, Y \rightarrow Z\}$. Decompose R into $R_1(X, Y)$ and $R_2(Y, Z)$. How many functional dependencies from the minimal cover of F are preserved?

Answer: 2 **Solution:** $X \rightarrow Y$ is directly checked on R_1 , and $Y \rightarrow Z$ is directly checked on R_2 . Both dependencies in the minimal cover are preserved. Count = 2.

Q3.9

[GATE CS 2017 | MCQ | 1 Mark]

Why is 3NF often preferred over BCNF in practical relational database design?

- (A) 3NF always guarantees preservation of all functional dependencies while maintaining a lossless join, whereas BCNF may lose dependencies.
- (B) 3NF tables take less disk space than BCNF tables.

- (C) 3NF relations have no update anomalies.
- (D) BCNF cannot handle foreign key constraints.

Answer: (A) **Solution:** Achieving BCNF sometimes requires sacrificing dependency preservation, meaning enforcing constraints requires expensive inter-relational joins. 3NF always guarantees both lossless join and dependency preservation.

Q3.10

[GATE DA 2026 | NAT | 2 Marks]

Relation $R(A, B, C)$ has functional dependencies $F = \{AB \rightarrow C, C \rightarrow B\}$. What is the number of attributes in the intersection of candidate keys? **Answer: 1** **Solution:** Candidate keys are AB and AC . The intersection of candidate keys is $\{A, B\} \cap \{A, C\} = \{A\}$. The number of attributes in the intersection is 1.

3.2 10 Practice Questions (GATE Level)

P3.1

[Practice | MCQ | 1 Mark]

Consider relation $R(A, B, C, D)$ with functional dependency set $F = \{A \rightarrow BCD, BC \rightarrow AD\}$. Which of the following is TRUE?

- (A) R is in BCNF.
- (B) R is in 3NF but not in BCNF.
- (C) R is in 2NF but not in 3NF.
- (D) R is in 1NF only.

Answer: (A) **Solution:** $A^+ = \{A, B, C, D\} \implies A$ is a candidate key. $(BC)^+ = \{A, B, C, D\} \implies BC$ is a candidate key. Both determinants A and BC are superkeys. Thus, R is in BCNF.

P3.2

[Practice | NAT | 2 Marks]

Relation $R(A, B, C, D)$ has $F = \{AB \rightarrow C, C \rightarrow D, D \rightarrow A\}$. If candidate keys are AB, BC, BD , how many functional dependencies in F violate BCNF? **Answer: 2** **Solution:** Superkeys are supersets of AB, BC, BD . Testing each dependency:

1. $AB \rightarrow C$: AB is a candidate key (superkey), so it does not violate BCNF.
 2. $C \rightarrow D$: $C^+ = \{C, D, A\} \neq \{A, B, C, D\}$, so C is not a superkey. Violates BCNF.
 3. $D \rightarrow A$: $D^+ = \{D, A\} \neq \{A, B, C, D\}$, so D is not a superkey. Violates BCNF.
- Thus, exactly 2 dependencies violate BCNF.

P3.3

[Practice | MSQ | 2 Marks]

Relation $R(A, B, C)$ has dependencies $F = \{AB \rightarrow C, C \rightarrow A\}$. Consider decomposing R into $R_1(A, C)$ and $R_2(B, C)$. Which of the following properties hold?

- (A) The decomposition is lossless.
- (B) The decomposition is dependency preserving.
- (C) Both R_1 and R_2 are in BCNF.
- (D) The dependency $AB \rightarrow C$ is preserved.

Answer: (A, C) **Solution:** (A) $R_1 \cap R_2 = \{C\}$. In R_1 , $C \rightarrow A$ holds, so C is a key of R_1 , making the decomposition lossless. (C) In $R_1(A, C)$, $C \rightarrow A$ holds with key C (in BCNF); in $R_2(B, C)$, only trivial dependencies hold (in BCNF). (B) and (D) are false because $AB \rightarrow C$ involves attributes split across tables and cannot be derived from $F_1 \cup F_2 = \{C \rightarrow A\}$.

P3.4

[Practice | NAT | 1 Mark]

A relation $R(A, B)$ has two attributes. What is the maximum number of functional dependencies in its minimal cover that can cause a violation of BCNF? **Answer: 0** **Solution:** Every binary relation $R(A, B)$ is inherently in BCNF. The only possible non-trivial dependencies are $A \rightarrow B$, $B \rightarrow A$, or both, each of which makes the determinant a candidate key. No dependency can violate BCNF.

P3.5

[Practice | MCQ | 2 Marks]

Let $R(A, B, C, D, E)$ have candidate key A and functional dependencies $F = \{A \rightarrow BCDE, B \rightarrow C, D \rightarrow E\}$. To decompose R into BCNF, what is the minimum number of relations required?

- (A) 2
- (B) 3
- (C) 4
- (D) 5

Answer: (B) **Solution:** Candidate key is A . $B \rightarrow C$ violates BCNF (B not superkey). Decompose into $R_1(B, C)$ (in BCNF) and $R'(A, B, D, E)$. In R' , key is A , and $D \rightarrow E$ violates BCNF (D not superkey). Decompose R' into $R_2(D, E)$ (in BCNF) and $R_3(A, B, D)$ (in BCNF, key A). Total relations = 3.

P3.6

[Practice | NAT | 2 Marks]

Decomposition of $R(A, B, C, D)$ into $R_1(A, B, C)$ and $R_2(C, D)$ is performed under $F = \{A \rightarrow B, B \rightarrow C, C \rightarrow D\}$. Is the common attribute set a candidate key for R_2 ? Enter 1 for Yes, 0 for No.

Answer: 1 **Solution:** $R_1 \cap R_2 = \{C\}$. In R_2 , $C \rightarrow D$ holds, so $C^+ = \{C, D\} = R_2$. Thus C is a candidate key for R_2 .

P3.7

[Practice | MCQ | 1 Mark]

If testing for dependency preservation in a decomposition $D = \{R_1, R_2, \dots, R_k\}$, a dependency $X \rightarrow Y \in F$ is preserved if:

- (A) $X \rightarrow Y \in (F_1 \cup F_2 \cup \dots \cup F_k)^+$
- (B) $X \cup Y \subseteq R_i$ for some single relation R_i
- (C) $(R_1 \cap R_2) \rightarrow X$
- (D) X is a candidate key of all R_i

Answer: (A) **Solution:** Dependency preservation requires that the union of the projected functional dependencies $(F_1 \cup \dots \cup F_k)$ logically implies every dependency in the original set F , i.e., $F^+ = (F_1 \cup \dots \cup F_k)^+$.

P3.8

[Practice | MSQ | 2 Marks]

Which of the following conditions is/are sufficient to conclude that a relation R is in BCNF?

- (A) All attributes of R together form the only candidate key.
- (B) R is in 3NF and all candidate keys are single (simple) attributes.
- (C) R contains only two attributes and is in 1NF.
- (D) Every attribute of R is prime.

Answer: (A, C) **Solution:** (A) If the only candidate key is all attributes, any non-trivial dependency must have all attributes on the LHS, making it a superkey. (C) Every 2-attribute relation is in BCNF. (D) is false: e.g., $R(A, B, C)$ with $AB \rightarrow C, C \rightarrow B$ has all prime attributes and is in 3NF, but violates BCNF. (B) is false if keys overlap.

P3.9**[Practice | NAT | 1 Mark]**

Given relation $R(A, B, C)$ with $F = \{A \rightarrow B, B \rightarrow C, C \rightarrow A\}$. How many candidate keys are there? **Answer: 3** **Solution:** $A^+ = \{A, B, C\}$, $B^+ = \{A, B, C\}$, and $C^+ = \{A, B, C\}$. Each single attribute is a candidate key. Total = 3.

P3.10**[Practice | MCQ | 1 Mark]**

When a relation is decomposed into BCNF, which of the following is GUARANTEED to hold?

- (A) Lossless join only
- (B) Dependency preservation only
- (C) Both lossless join and dependency preservation
- (D) Neither lossless join nor dependency preservation

Answer: (A) **Solution:** The standard BCNF decomposition algorithm guarantees a lossless join. However, dependency preservation cannot be guaranteed in all cases.

4 Topic 4: Mensuration & Geometry Basics

4.1 10 Previous Year Questions (PYQs)

Q4.1

[GATE GA 2024 (DA/CS) | MCQ | 1 Mark]

The radius of a circle is increased by 20%. By what percentage does the area of the circle increase?

- (A) 20%
- (B) 40%
- (C) 44%
- (D) 24%

Answer: (C) **Solution:** Area $A = \pi r^2$. New radius $r' = 1.20r$. New area $A' = \pi(1.20r)^2 = 1.44\pi r^2 = 1.44A$. The percentage increase is $(1.44 - 1) \times 100\% = 44\%$.

Q4.2

[GATE GA 2023 | NAT | 2 Marks]

A solid metallic sphere of radius 6 cm is melted and recast into small solid spheres of radius 2 cm each. How many small spheres are formed? **Answer: 27** **Solution:** Volume of a sphere is

$$V = \frac{4}{3}\pi r^3. \text{ Number of spheres } N = \frac{V_{\text{large}}}{V_{\text{small}}} = \frac{\frac{4}{3}\pi(6)^3}{\frac{4}{3}\pi(2)^3} = \left(\frac{6}{2}\right)^3 = 3^3 = 27.$$

Q4.3

[GATE GA 2022 | MCQ | 2 Marks]

In a right-angled triangle, the lengths of the two sides containing the right angle are 6 cm and 8 cm. What is the radius of the incircle (inradius) of the triangle?

- (A) 2 cm
- (B) 3 cm
- (C) 4 cm
- (D) 1.5 cm

Answer: (A) **Solution:** Hypotenuse $c = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = 10$ cm. Semi-perimeter $s = \frac{6+8+10}{2} = 12$ cm. Area $\Delta = \frac{1}{2} \times 6 \times 8 = 24$ cm². Inradius $r = \frac{\Delta}{s} = \frac{24}{12} = 2$ cm. (Alternatively, for a right-angled triangle, $r = \frac{a+b-c}{2} = \frac{6+8-10}{2} = \frac{4}{2} = 2$ cm.)

Q4.4

[GATE GA 2021 | NAT | 1 Mark]

A rectangular plot has length 30 m and breadth 20 m. A path of uniform width 2 m runs all around the outside of the plot. What is the area of the path in m²? **Answer: 216** **Solution:** Outer length = $30 + 2(2) = 34$ m. Outer breadth = $20 + 2(2) = 24$ m. Outer area = $34 \times 24 = 816$ m². Inner area = $30 \times 20 = 600$ m². Area of the path = $816 - 600 = 216$ m².

Q4.5

[GATE GA 2024 | MCQ | 1 Mark]

If the diagonal of a square is $10\sqrt{2}$ cm, what is its perimeter?

- (A) 20 cm
- (B) 40 cm
- (C) 30 cm
- (D) $40\sqrt{2}$ cm

Answer: (B) **Solution:** Diagonal of a square $d = a\sqrt{2} \implies 10\sqrt{2} = a\sqrt{2} \implies a = 10$ cm. Perimeter $P = 4a = 4 \times 10 = 40$ cm.

Q4.6

[GATE GA 2020 | NAT | 2 Marks]

A right circular cylinder of height 14 cm has a curved surface area of 176 cm^2 . Taking $\pi = \frac{22}{7}$, what is the diameter of the base of the cylinder in cm? **Answer: 4** **Solution:** Curved surface area $= 2\pi rh$. Thus, $2 \times \frac{22}{7} \times r \times 14 = 176 \implies 88r = 176 \implies r = 2$ cm. The base diameter is $2r = 4$ cm.

Q4.7

[GATE GA 2019 | MCQ | 2 Marks]

An equilateral triangle and a regular hexagon have equal perimeters. What is the ratio of the area of the triangle to the area of the hexagon?

- (A) 2 : 3
- (B) 3 : 4
- (C) 1 : 2
- (D) 4 : 9

Answer: (A) **Solution:** Let common perimeter be $6s$. Side of equilateral triangle $a = \frac{6s}{3} = 2s$. Side of regular hexagon $b = \frac{6s}{6} = s$. Area of triangle $= \frac{\sqrt{3}}{4}a^2 = \frac{\sqrt{3}}{4}(2s)^2 = \sqrt{3}s^2$. Area of regular hexagon $= 6 \times \left(\frac{\sqrt{3}}{4}b^2\right) = \frac{6\sqrt{3}}{4}s^2 = \frac{3\sqrt{3}}{2}s^2$. Ratio $= \frac{\sqrt{3}s^2}{\frac{3\sqrt{3}}{2}s^2} = \frac{1}{3/2} = \frac{2}{3}$.

Q4.8

[GATE GA 2025 | NAT | 2 Marks]

A right circular cone of base radius 5 cm has a slant height of 13 cm. Taking $\pi = \frac{22}{7}$, what is the volume of the cone in cm^3 ? (Round off to two decimal places) **Answer: 314.29** **Solution:** Height $h = \sqrt{l^2 - r^2} = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$ cm. Volume $V = \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times 25 \times 12 = 100 \times \frac{22}{7} = \frac{2200}{7} \approx 314.29 \text{ cm}^3$.

Q4.9

[GATE GA 2023 | MCQ | 1 Mark]

The interior angle of a regular polygon is 144° . How many sides does the polygon have?

- (A) 8
- (B) 9
- (C) 10
- (D) 12

Answer: (C) **Solution:** Each exterior angle is $180^\circ - 144^\circ = 36^\circ$. The sum of exterior angles of any convex polygon is 360° . Number of sides $n = \frac{360^\circ}{36^\circ} = 10$.

Q4.10

[GATE GA 2026 | NAT | 1 Mark]

The ratio of the areas of two similar triangles is 16 : 25. If the altitude of the smaller triangle is 8 cm, what is the corresponding altitude of the larger triangle in cm? **Answer: 10** **Solution:** For similar triangles, the ratio of areas equals the ratio of squares of corresponding linear dimensions: $\frac{A_1}{A_2} = \left(\frac{h_1}{h_2}\right)^2 \implies \frac{16}{25} = \left(\frac{8}{h_2}\right)^2 \implies \frac{4}{5} = \frac{8}{h_2} \implies h_2 = \frac{8 \times 5}{4} = 10$ cm.

4.2 10 Practice Questions (GATE Level)

P4.1

[Practice | MCQ | 1 Mark]

If the length of a rectangle is increased by 30% and the breadth is decreased by 20%, what is the net percentage change in the area?

- (A) Increases by 10%
- (B) Increases by 4%
- (C) Decreases by 4%
- (D) Increases by 6%

Answer: (B) **Solution:** Net change formula: $a + b + \frac{ab}{100} = 30 + (-20) + \frac{(30)(-20)}{100} = 10 - 6 = +4\%$. The area increases by 4%.

P4.2

[Practice | NAT | 2 Marks]

A wire is bent into the shape of a circle of radius 28 cm. If the same wire is rebent into the shape of a square, what is the side length of the square in cm? (Take $\pi = \frac{22}{7}$) **Answer: 44** **Solution:** Circumference of circle $C = 2\pi r = 2 \times \frac{22}{7} \times 28 = 2 \times 22 \times 4 = 176$ cm. Perimeter of the square is $4a = 176 \implies a = \frac{176}{4} = 44$ cm.

P4.3

[Practice | MSQ | 2 Marks]

Which of the following statements about three-dimensional solid geometries is/are TRUE?

- (A) For a given fixed surface area, a sphere encloses the maximum possible volume among all 3D solids.
- (B) The total surface area of a hemisphere of radius r including its flat circular base is $3\pi r^2$.
- (C) If all linear dimensions of a cuboid are doubled, its volume increases by a factor of 8.
- (D) The volume of a right circular cone is half the volume of a cylinder with the same base radius and height.

Answer: (A, B, C) **Solution:** (A) Follows from the 3D isoperimetric inequality. (B) Curved surface $2\pi r^2 + \text{base } \pi r^2 = 3\pi r^2$. (C) Scaling factor is $2^3 = 8$. (D) is false; cone volume is $\frac{1}{3}\pi r^2 h$, which is one-third (not half) of the cylinder volume.

P4.4

[Practice | NAT | 2 Marks]

The perimeter of a rhombus is 40 cm, and the length of one of its diagonals is 12 cm. What is the area of the rhombus in cm^2 ? **Answer: 96** **Solution:** Side of rhombus $a = \frac{40}{4} = 10$ cm. Diagonals of a rhombus bisect each other at right angles. Half-diagonal $d_1/2 = 6$ cm. The other half-diagonal is $\sqrt{10^2 - 6^2} = \sqrt{100 - 36} = \sqrt{64} = 8$ cm $\implies d_2 = 16$ cm. Area $= \frac{1}{2}d_1d_2 = \frac{1}{2} \times 12 \times 16 = 96 \text{ cm}^2$.

P4.5

[Practice | MCQ | 1 Mark]

A solid cylinder has base diameter 14 cm and height 10 cm. What is its total surface area in cm^2 ? (Take $\pi = \frac{22}{7}$)

- (A) 748
- (B) 440
- (C) 594
- (D) 880

Answer: (A) **Solution:** Radius $r = 7$ cm, height $h = 10$ cm. Total surface area $= 2\pi r(h + r) = 2 \times \frac{22}{7} \times 7 \times (10 + 7) = 44 \times 17 = 748 \text{ cm}^2$.

P4.6

[Practice | NAT | 2 Marks]

In $\triangle ABC$, $DE \parallel BC$ such that D lies on AB and E lies on AC . If $AD = 3$ cm, $DB = 5$ cm, and the area of $\triangle ADE$ is 18 cm^2 , what is the area of the quadrilateral $BDEC$ in cm^2 ? **Answer: 110**

Solution: $\triangle ADE \sim \triangle ABC$. Linear ratio is $\frac{AD}{AB} = \frac{3}{3+5} = \frac{3}{8}$. Ratio of areas: $\frac{\text{Area}(\triangle ADE)}{\text{Area}(\triangle ABC)} = \left(\frac{3}{8}\right)^2 = \frac{9}{64}$.

Given $\text{Area}(\triangle ADE) = 18$, we have $\frac{18}{\text{Area}(\triangle ABC)} = \frac{9}{64} \implies \text{Area}(\triangle ABC) = \frac{18 \times 64}{9} = 128 \text{ cm}^2$. Area of quadrilateral $BDEC = 128 - 18 = 110 \text{ cm}^2$.

P4.7

[Practice | MCQ | 1 Mark]

What is the length of the longest rod that can be placed inside a rectangular room of dimensions $10 \text{ m} \times 10 \text{ m} \times 5 \text{ m}$?

- (A) 15 m
- (B) 12.5 m
- (C) 20 m
- (D) 18 m

Answer: (A) **Solution:** The space diagonal of a cuboid is $D = \sqrt{l^2 + w^2 + h^2} = \sqrt{10^2 + 10^2 + 5^2} = \sqrt{100 + 100 + 25} = \sqrt{225} = 15 \text{ m}$.

P4.8

[Practice | NAT | 2 Marks]

A sector of a circle of radius 12 cm has a central angle of 60° . What is the perimeter of the sector in cm? (Take $\pi = 3.14$) (Round off to two decimal places) **Answer: 36.56** **Solution:** Arc length $L = \frac{\theta}{360^\circ} \times 2\pi r = \frac{60}{360} \times 2 \times 3.14 \times 12 = \frac{1}{6} \times 75.36 = 12.56 \text{ cm}$. Perimeter includes the two bounding radii: $P = L + 2r = 12.56 + 2(12) = 12.56 + 24 = 36.56 \text{ cm}$.

P4.9

[Practice | MSQ | 2 Marks]

Let a circle be inscribed in a square of side length $2a$, and another circle circumscribed around the same square. Which of the following statements is/are TRUE?

- (A) The radius of the inscribed circle is a .
- (B) The radius of the circumscribed circle is $a\sqrt{2}$.
- (C) The ratio of the area of the inscribed circle to the area of the circumscribed circle is $1 : 2$.
- (D) The area of the square is $\frac{4}{\pi}$ times the area of the inscribed circle.

Answer: (A, B, C, D) **Solution:** Side of square $= 2a$. (A) Inradius $r = \frac{2a}{2} = a$. (B) Diagonal $= 2a\sqrt{2}$, so circumradius $R = \frac{2a\sqrt{2}}{2} = a\sqrt{2}$. (C) Ratio of circle areas $= \frac{\pi a^2}{\pi (a\sqrt{2})^2} = \frac{a^2}{2a^2} = \frac{1}{2}$. (D) Area of square $= (2a)^2 = 4a^2$. Area of inscribed circle $= \pi a^2$. The ratio is $\frac{4a^2}{\pi a^2} = \frac{4}{\pi}$.

P4.10

[Practice | NAT | 1 Mark]

How many diagonals does a regular convex polygon with 8 vertices (an octagon) have? **Answer: 20** **Solution:** The number of diagonals in an n -sided polygon is $\frac{n(n-3)}{2}$. For $n = 8$, this is $\frac{8(8-3)}{2} = \frac{8 \times 5}{2} = 20$.